Pritam Talukdar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.92
Maharishi Vidya Mandir, Barsajai	Class XII (CBSE)	2018	88.80%
Maharishi Vidya Mandir, Barsajai	Class X (CBSE)	2016	10.00%

M.TECH THESIS

Experimental and Numerical Investigation of Open-Hole Carbon Fiber Reinforced Polymer (CFRP) Laminates under combined Tension-Shear Fatigue Loading

- Fabricated quasi-isotropic and double-double **CFRP laminates** and prepared 60 open-hole specimens with bonded GFRP tabs for combined tension-shear static and **fatigue testing** using an in-house developed Modified Arcan Fixture.
- Integrated a 100 kN Universal Testing Machine (UTM), 3D Digital Image Correlation (DIC) and Acoustic Emission (AE) to study deformation, **stiffness degradation** and **damage progression** during **fatigue** loading in **composites**.
- Developing **Abaqus finite element models** with the LARC05 damage criterion and **Cohesive Zone Modelling** (CZM) to study **damage initiation**, **failure** and **delamination**, correlating numerical predictions with experimental results.

WORK EXPERIENCE

Project Associate, Council of Scientific and Industrial Research (CSIR-CSIO), Chandigarh

- Worked on **new product development** and **reverse engineering** of mechanical and opto-mechanical **aerospace components**.
- Developed **3D CAD models** for **components** and designed **fixtures** to support **qualification tests** across multiple projects.
- Prepared **2D manufacturing drawings** with **GD&T** for components and coordinated with **external stakeholders** through **meetings, design reviews** and **project documentation**.

GEA, Defence Research and Development Organisation (DRDO), Balasore

- Developed **3D CAD models** of mechanical components and projectiles in **SolidWorks** for defence-related engineering work.
- Conducted field **data acquisition** using DAQ systems, strain gauges, microphones, and other sensors during testing activities.
- Assisted scientists in various technical and experimental activities, supporting field trials, data collection and analysis.

SKILLS

CAE/FEA: Abaqus, Ansys, HyperMesh

CAD: SolidWorks, NX, 2D Manufacturing Drawings, GD&T

Core Engineering: New Product Development, Research & Development, Product Design and Testing, Technical Documentation

Experimental: Composite Fabrication, Universal Testing Machine, Digital Image Correlation, Acoustic Emission Sensors

Mechanics & Testing: Composite Mechanics, Static & Fatigue Testing, Fracture Mechanics, Multiaxial Loading

Programming: MATLAB, Python

PROJECTS

New Product Development of Frangible Crash-Detection Switch for Do-228 Aircraft | *CSIR*

- **Led a 3-member team** in **design and development** of a frangible crash-detection switch for **Hindustan Aeronautics Limited (HAL), Korwa**, designed for the **Do-228 aircraft** to trigger the Deployable **Digital Flight Data Recorder (DFDR)** during a crash.
- Designed a **spring-loaded mechanical switching mechanism** using a **borosilicate glass envelope** and stainless-steel spring, converting crash-induced glass fracture into rapid electrical contact closure to trigger the DFDR release system.
- Engineered the spring-mass actuation mechanism for an estimated **0.29 ms** contact closure response against the specified <1 ms crash-response requirement, supporting rapid activation following severe impact.
- Developed the compact switch assembly within a **60x38x100 mm** envelope and 0.1 kg maximum mass, incorporating an anodized Ti-6Al-4V mounting flange for secure, corrosion-resistant integration with the **aircraft structure**.
- Designed for 8 **MIL-STD-810D** environmental **qualification tests**, including 20 g mechanical shock, 6-axis acceleration, random vibration, -55 °C to +85 °C thermal shock and 26 transit drops from 1.22 m.
- Coordinated with HAL Korwa through multiple site visits, meetings and **design reviews**, addressing **stakeholder requirements** and supporting technical documentation and project formalities.

Numerical & Experimental Validation of Progressive Damage in Open-Hole CFRP Laminates under Tension Shear Loading | *M.Tech*

- Developed a **finite element model** in Abaqus for open-hole **CFRP laminates** under combined tension shear static loading.
- Implemented **progressive damage** using the **LaRC05 failure criterion** and a **Cohesive Zone Model (CZM)** to capture intralaminar failure and interlaminar delamination.
- Validated experimental and numerical results against published results from Mahesh et al., comparing load displacement response, stiffness, damage initiation and failure behaviour.

Prediction of Geometric Correction Factor for a Single Crack under Uniaxial Loading using Automated Abaqus Simulations and Machine Learning | *M.Tech*

- **Automated** 100 fracture mechanics simulations in **ABAQUS** using **Python scripting**, sweeping crack length from 16 mm to 40.75 mm in 0.25 mm increments to extract the Mode-I Stress Intensity Factor (KI).
- Used quarter-point elements at the crack tip and averaged KI values from the 2nd to 5th contours to improve result stability.
- Built a fourth-order polynomial **regression model** predicting the geometry correction factor directly from the crack-length ratio, matching the theoretical solution to within 0.1% and eliminating the need for repeated Abaqus simulations.

Data-Driven Prediction of Mechanical Properties of Iron Alloys | *M.Tech*

- Built a **Python ML pipeline** to predict **UTS and Young's Modulus** of iron alloys using composition and material-property data.
- Compared **Linear Regression, Random Forest and Gradient Boosting** models using MSE, RMSE and R2 for final prediction.

Engine Sensor Analytics for Operational Mode Classification & Fuel-Efficiency Prediction | *M.Tech*

- Engineered **time-series, vibration and lag features** from multi-sensor engine data to capture operating-state variations.
- Developed **stacked classification and regression models** to jointly predict engine operating mode and fuel efficiency.

Machine Learning-Based Material Suitability Prediction for Electric Vehicle Chassis | *M.Tech*

- Engineered **strength, stiffness and material-ratio features** and removed redundant variables using feature importance.
- Built and tuned **Random Forest** and **ANN models** with cross-validation for material suitability prediction.

Design and Development of Projection Beam Combiners for Head-Up Display (HUD) of Jaguar Aircraft | *CSIR*

- **Performed reverse engineering** of projection beam combiners for the **Jaguar aircraft** for the indigenisation program of HAL Korwa, developing 3D CAD models based on existing components.
- **Prepared detailed 2D manufacturing drawings** with GD&T and manufacturing specifications for component fabrication.
- Coordinated component manufacturing and assembly, and designed **assembly and vibration fixtures** required for integration and testing of the beam combiner.
- Visited HAL Korwa for physical validation of the reverse-engineered model, checking hole-to-hole distances, dimensions, and component fit and participating in project meetings and maintaining required documentation.

Design and Development of Multi-Spectral AR-Coated Sapphire Fairing for IRST System | *CSIR*

- Designed the outer fairing assembly for the **Infrared Search and Track (IRST)** system of the **Su-30 aircraft** in **SolidWorks**, providing the enclosure for the internal sensing and tracking components.
- Modified the fairing design based on **industry requirements**, adding internal space for fan installation to provide cooling.
- Developed the complete assembly, integrating the modified components to ensure proper installation and operation.

INTERNSHIP

Winter Intern, Numaligarh Refinery Limited (NRL), Assam

- Studied the working principle of **gas turbines** in two 30 MW Gas Turbine Generators used for refinery power generation.
- Analysed the Heat Recovery Steam Generator (HRSG) and co-generation process, understanding how gas turbine exhaust heat is recovered for steam generation.

RELEVANT COURSES

Fracture Mechanics, Analysis and Design of Composite Structures, Finite Element Method, Advanced Finite Element Method, Advanced Mechanics of Solids, Machine Learning & Its Applications

CERTIFICATIONS

HyperMesh for FEA Pre-Processing Associate Specialist, Altair Learning and Education

Stress Analysis with Ansys Mechanical, Ansys

Six Sigma White Belt, Council for Six Sigma Certification (CSSC)

CONFERENCES

Poster Presentation: "*Reliability Analysis of an Opto- Mechanical Aircraft Crash Detection Device*" , OPTOIn 2024, CSIR-CSIO, Chandigarh

POSITIONS OF RESPONSIBILITY

Head Placement Coordinator of Mechanics & Design, Office of Career Services (OCS), IIT Hyderabad

- Reached out to **30+ companies** and coordinated with recruiters and OCS for placement activities.
- Coordinated placement activities through recruiter communication, student coordination, recruitment data tracking, and follow-ups.

Teaching Assistant, Advanced Mechanics of Solids & Finite Element Method

- Assisted faculty members in conducting academic activities across Advanced Mechanics of Solids and Finite Element Method.
- Evaluated assignments and examination answer scripts and assisted in examination conduct and invigilation for both subjects.

Sponsorship Team Member, MAE Department Day, IIT Hyderabad

- Assisted in managing departmental funds for the successful organization of MAE Department Day.
- Coordinated with the organizing team in planning and conducting various events and activities during the department day.

Volunteer, Office of Career Services (OCS), IIT Hyderabad

- Supported placement activities by monitoring online assessments and assisting with recruitment-related coordination.

Volunteer, OPTOIn 2024 International Conference

- Assisted in participant registration and coordinated with attendees during the conference.
- Supported invited speakers with transportation and other requirements, while assisting in the overall event management.

LANGUAGES

Japanese (Basic), English, Hindi, Assamese

HOBBIES & INTERESTS

Japanese Language & Culture, Fitness & Running, Travel, Photography, Cooking, Motorcycles & Formula 1, DIY Mechanical Repairs, Learning New Languages.